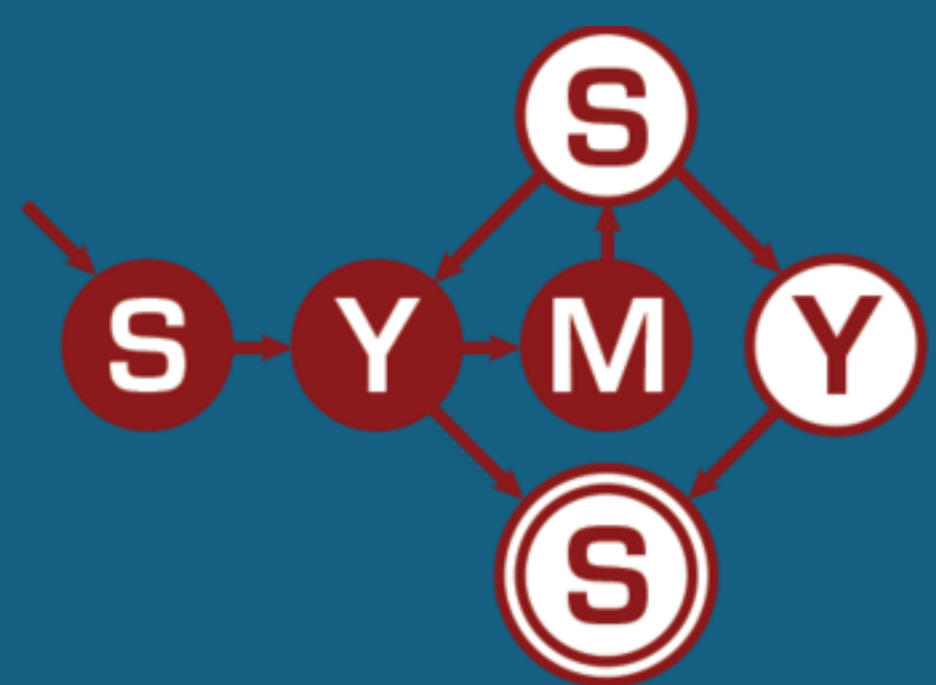
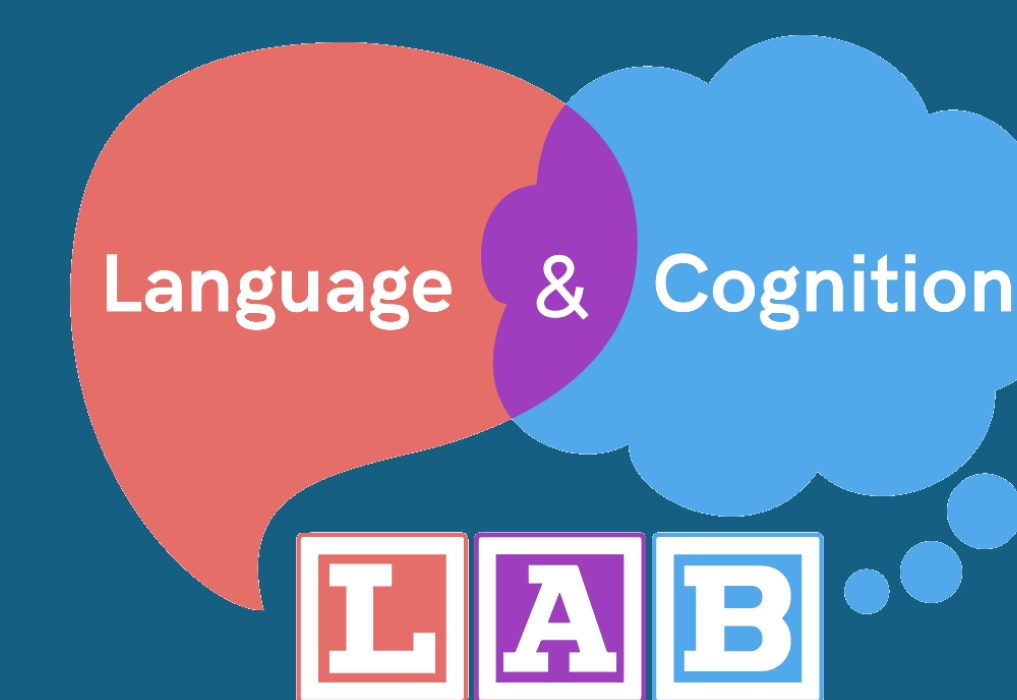




Investigating Children's Ability to Use Ad Hoc Referential Expressions

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Background

- Speakers use referential expressions to explain the characteristics of a referent in a manner that will prompt a listener to accurately identify the referent, even if both the speaker and listener have never seen it before.

- Example:

Can you grab my keys from inside?

Yeah, is it the ones with the yellow keychain and sparkly house key?

Yep! Those are mine.

- Adults use referential expressions all the time, seamlessly.
- Glucksberg, Krauss, & Weisberg published in 1966 that children aged 4 and 5 were unable to use referential expression.
- Their methodology was complicated, however, and likely not a good representation of kids' linguistic abilities.

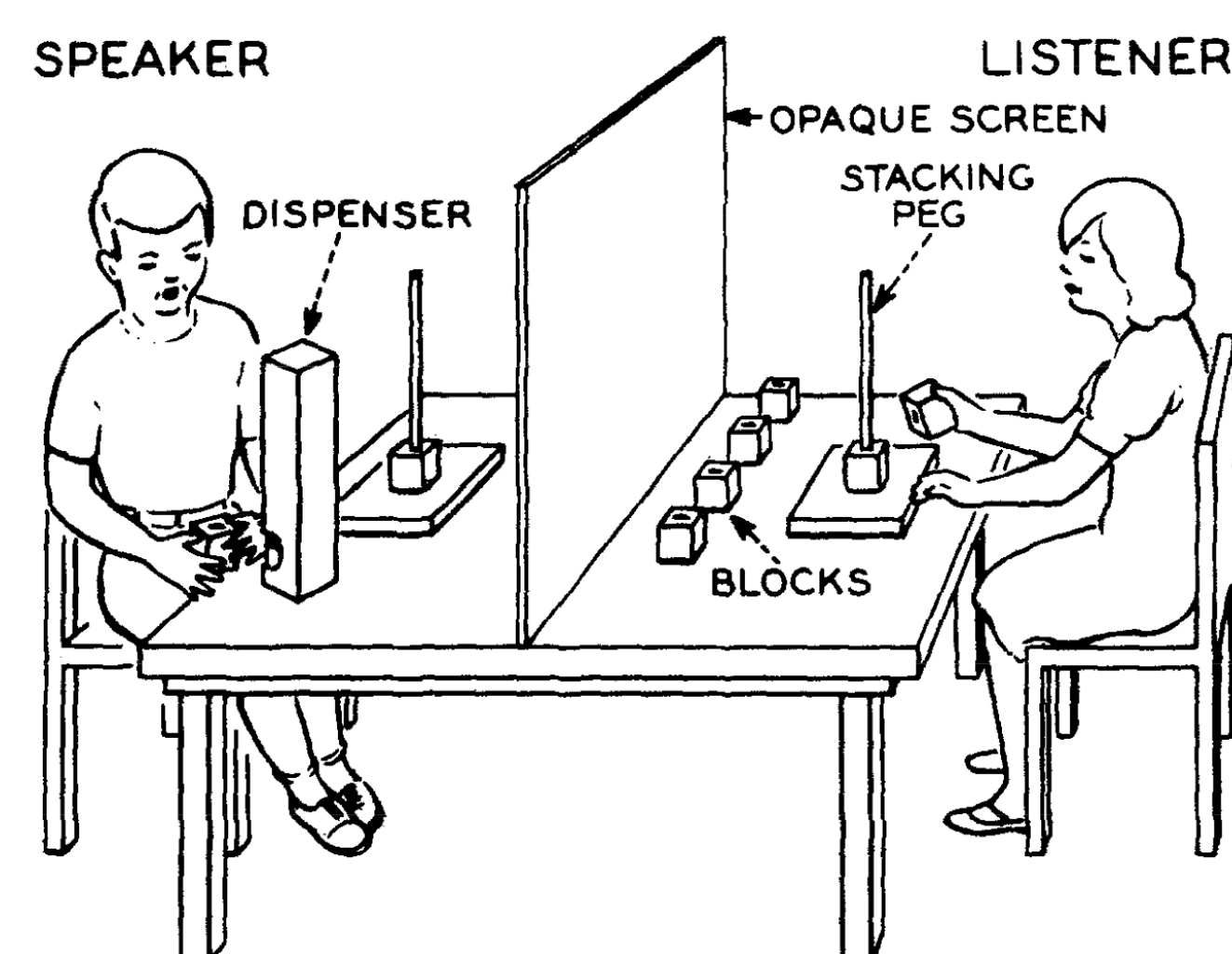


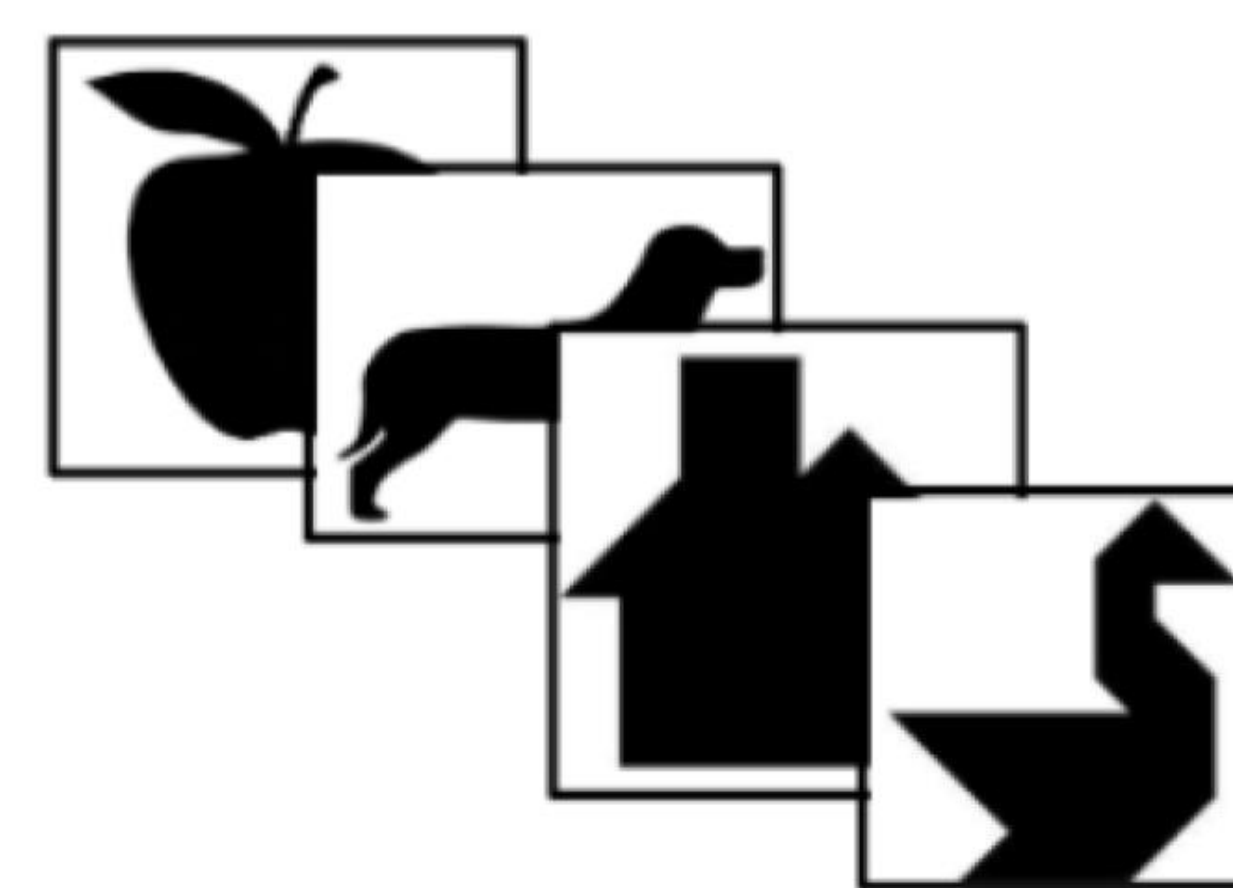
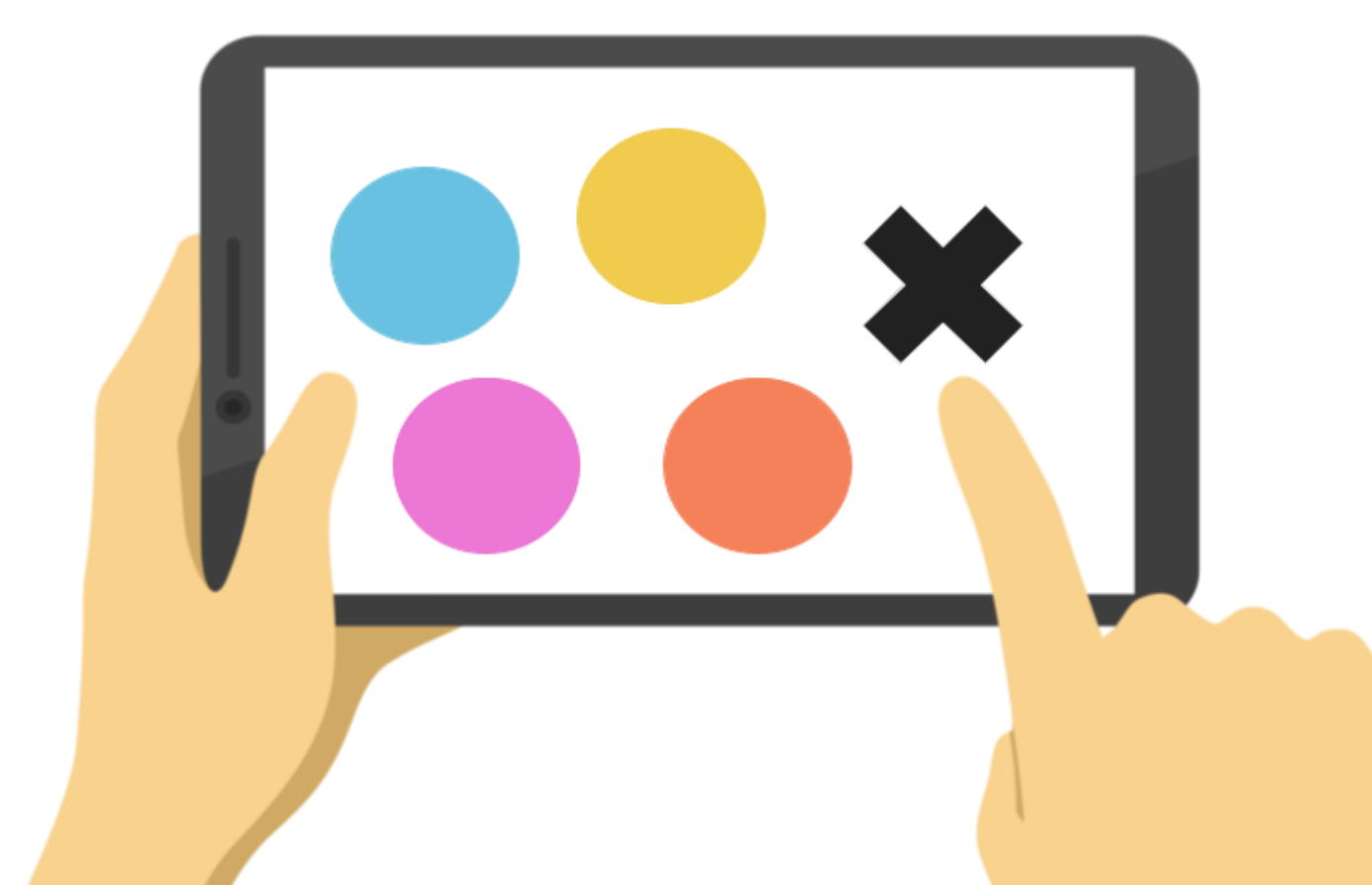
FIG. 2. The experimental task.

Glucksberg, et. al, methodology

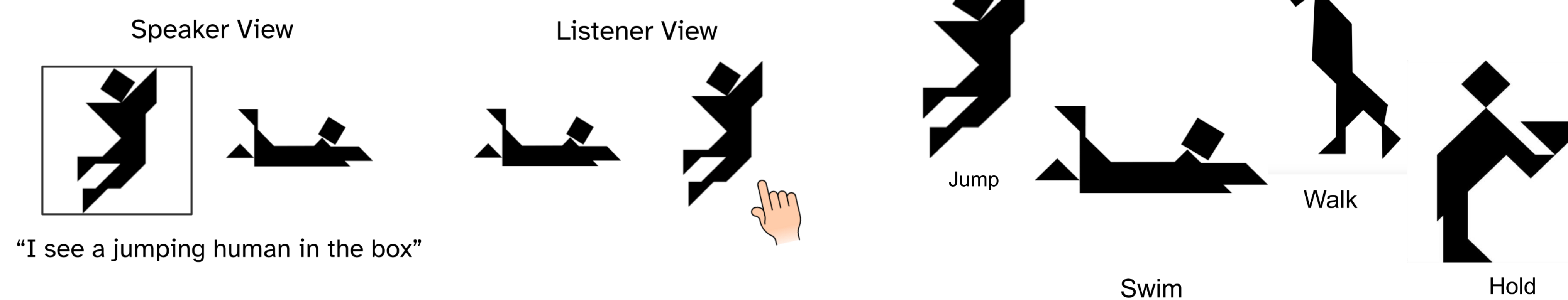
- We wanted to know if given a less complicated task to complete, can 4-to-5-year-old children form ad hoc referential expressions with each other to accurately describe novel referents?

Methods

- Participants are first given a bubble-popping task to acclimate them to touching the iPads before trials
- Afterwards, there are 4 warm up trials of increasing difficulty, to lesson the shock of describing the actual tangrams, which are more abstract and unrecognizable.

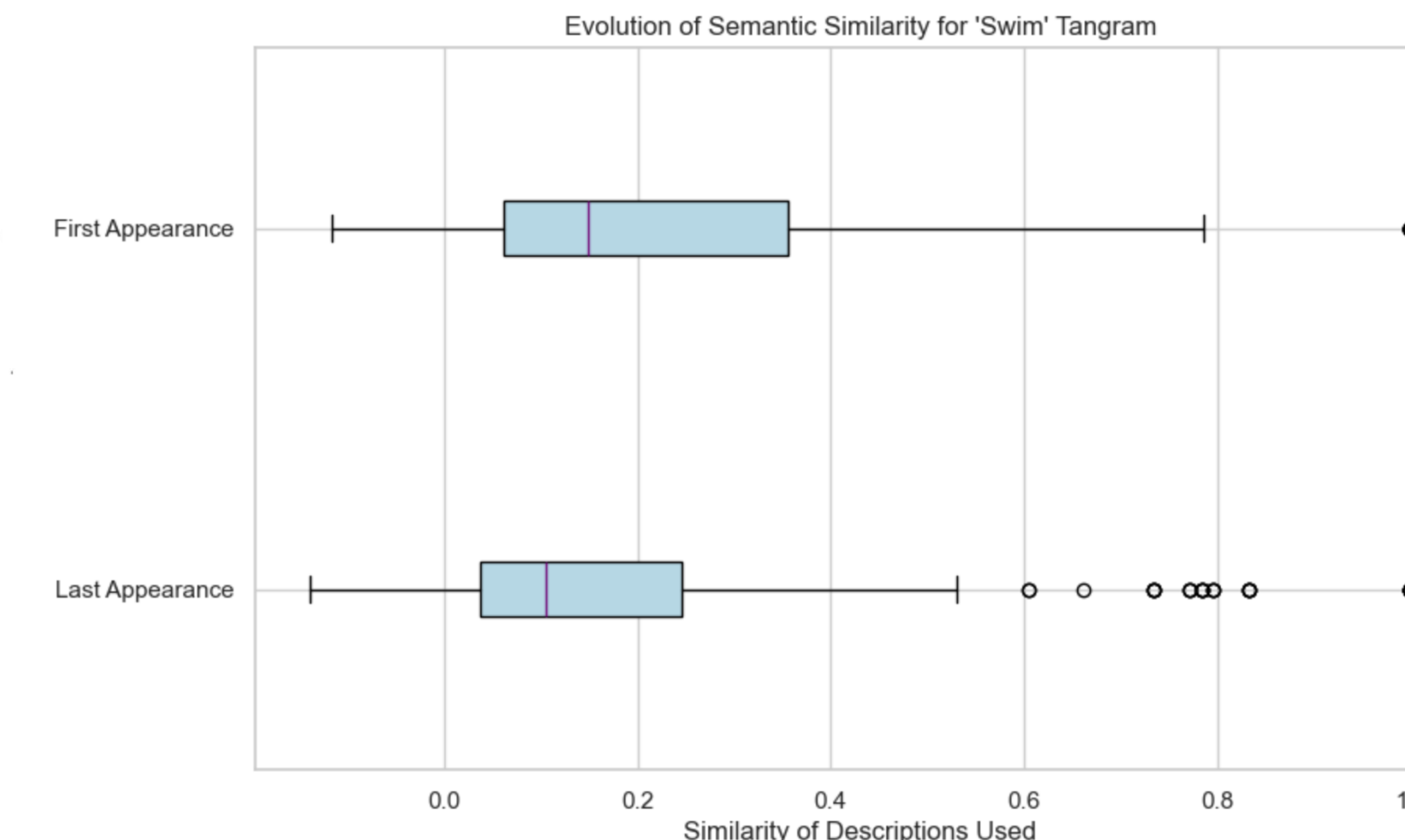
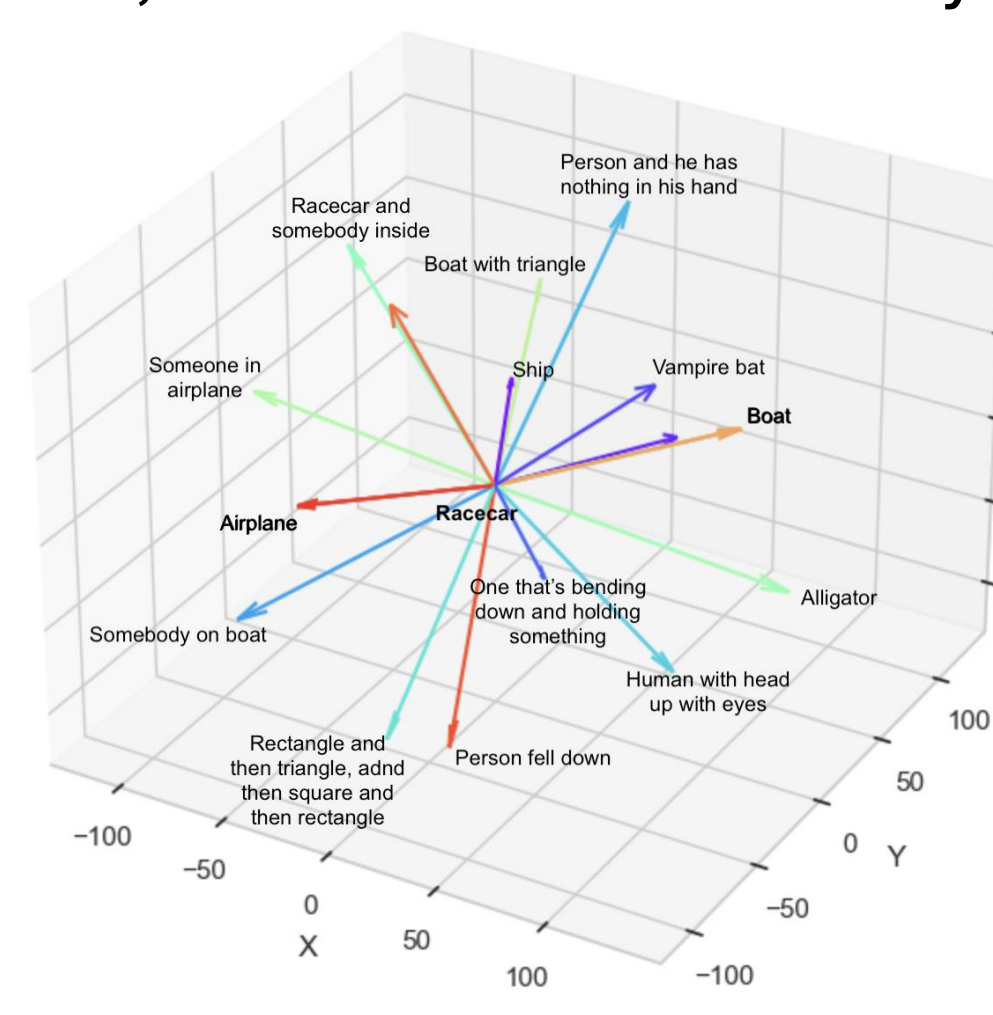
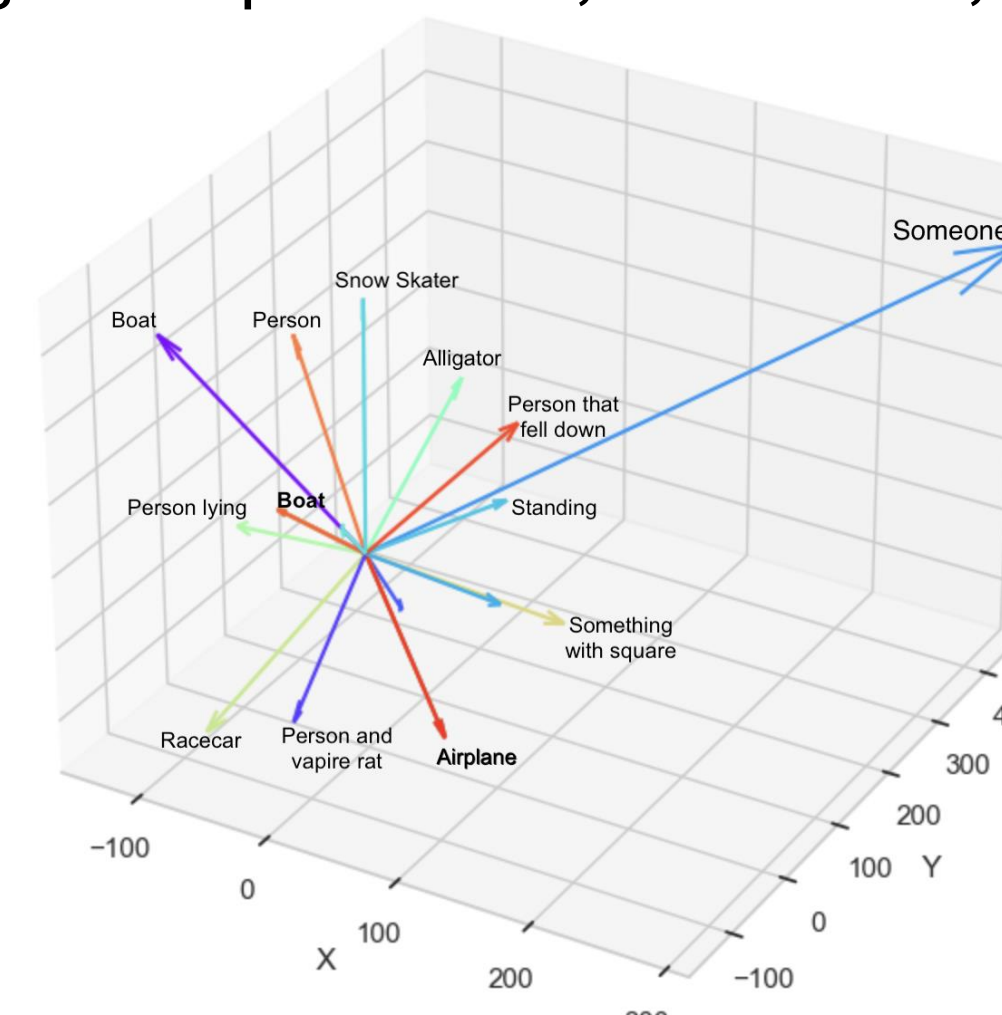


- Participants see two tangrams each on their iPads. One participant is the "speaker" who is instructed to communicate contents of boxed tangram to the other participant. The "listener" then must correctly tap the tangram being described to them. Participants switch roles every trial for 16 trials. Each of the 4 tangrams (jump, swim, walk, and hold) are tested 4 times.

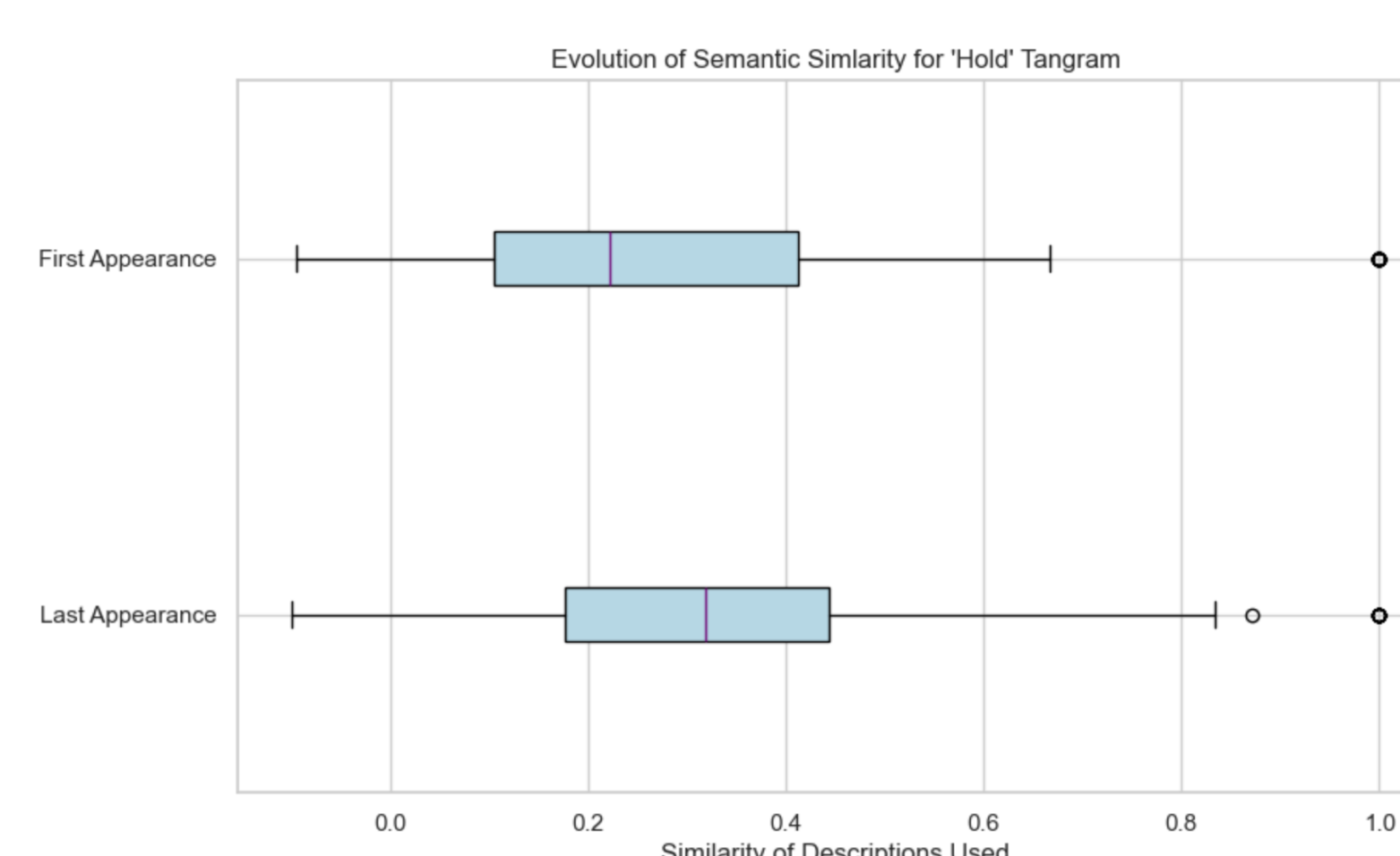
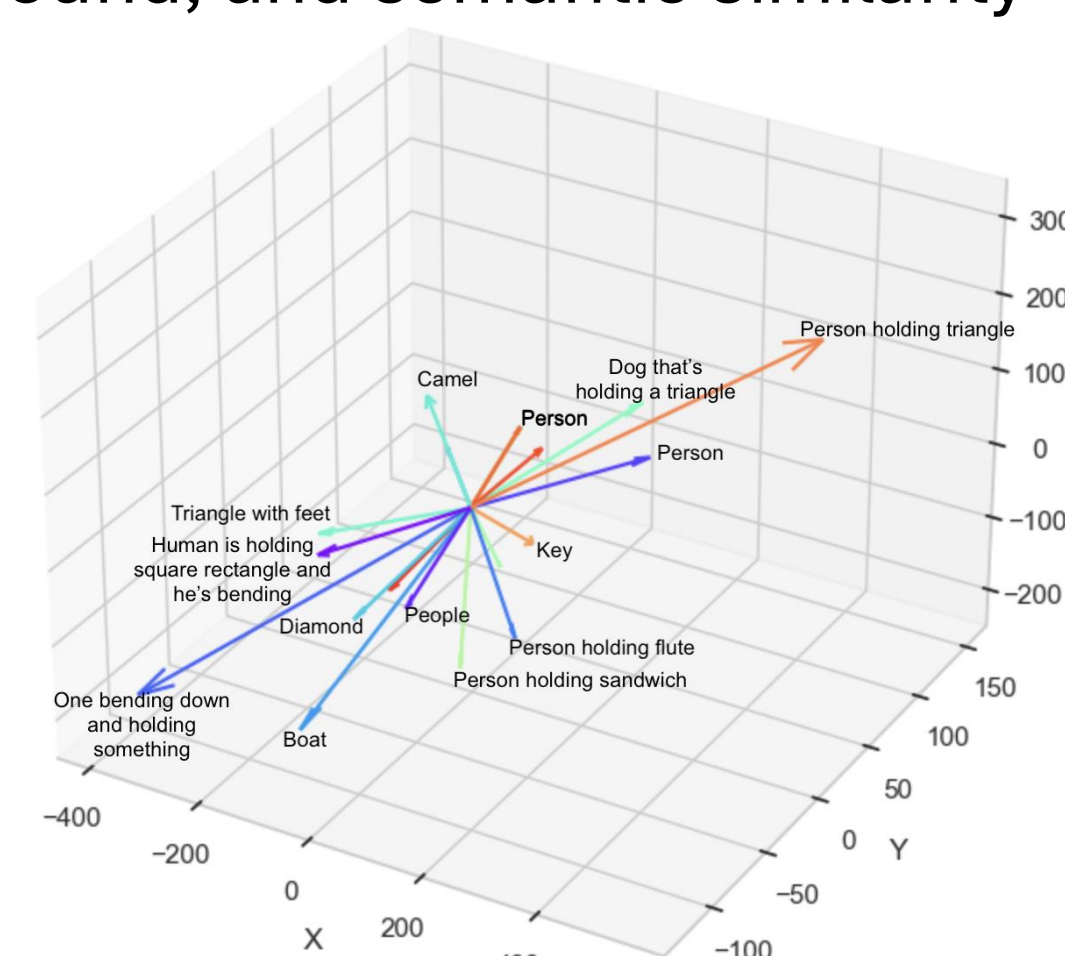
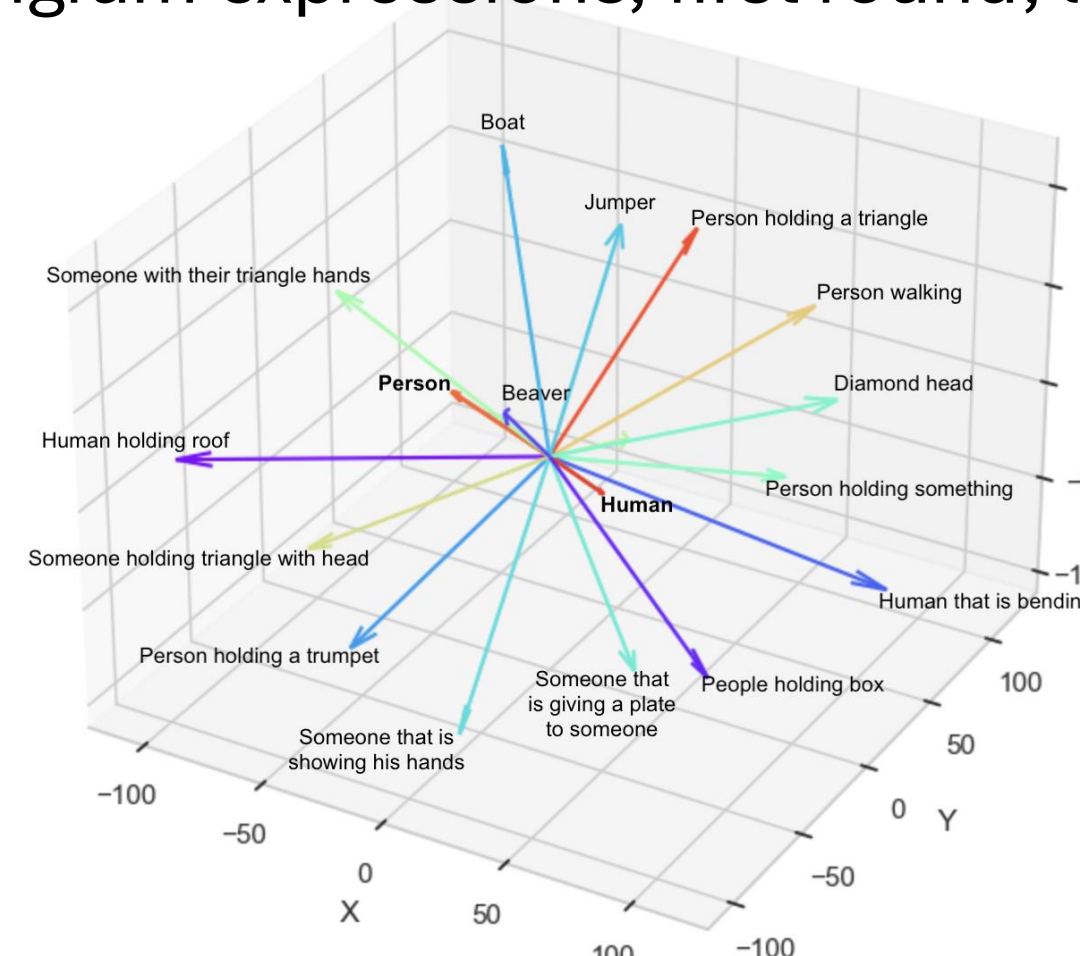


Results

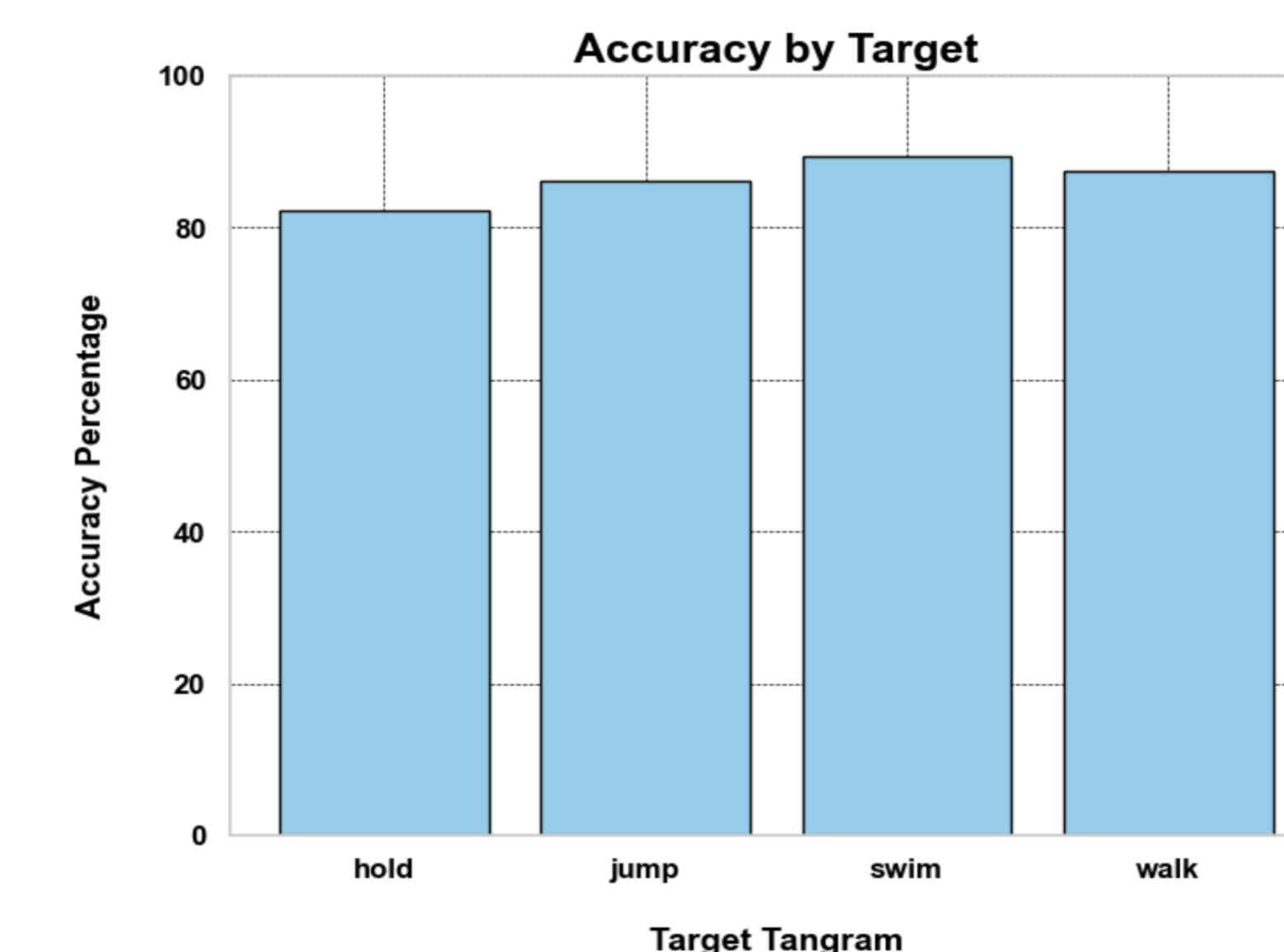
Swim tangram expressions, first round, last round, and semantic similarity



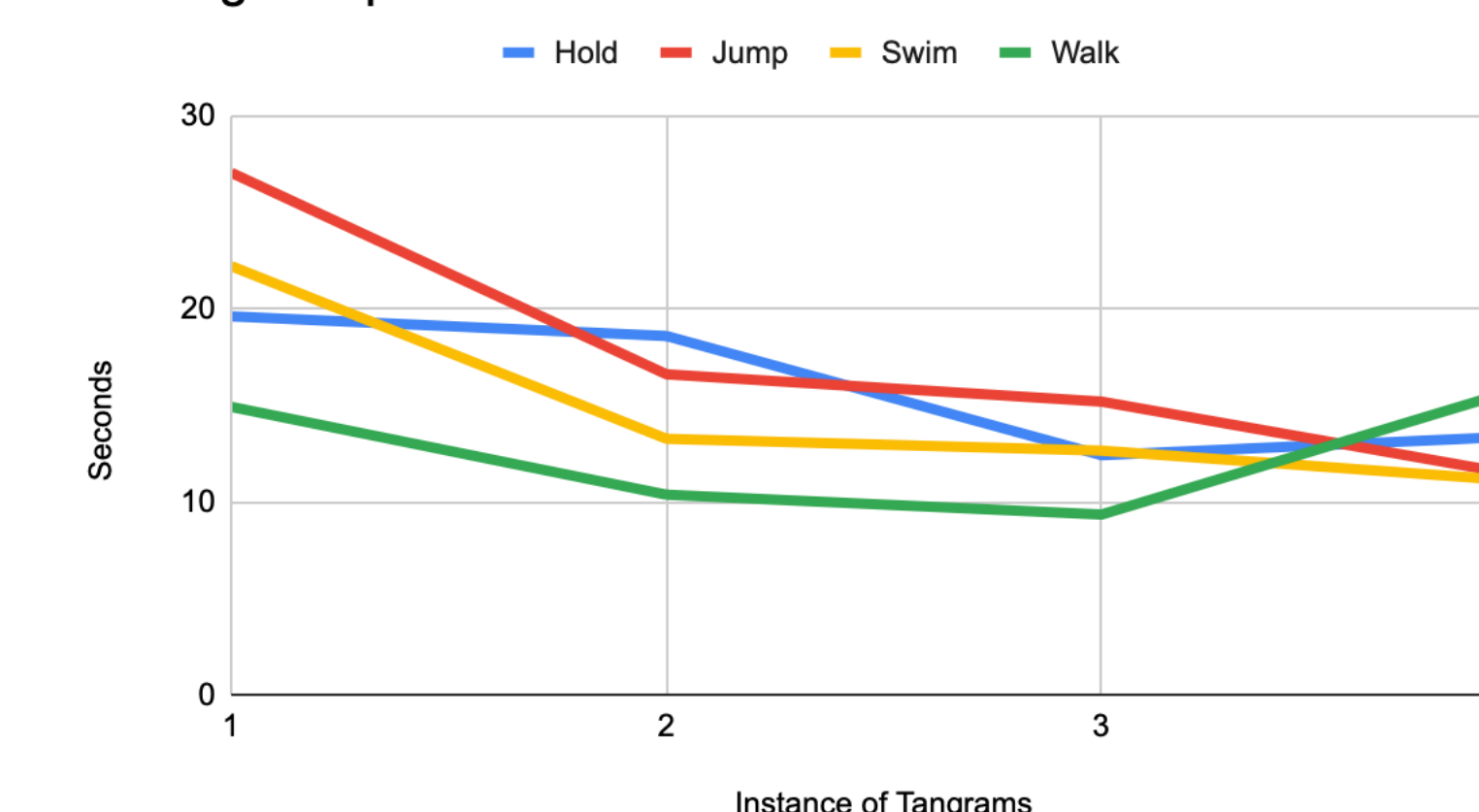
Hold tangram expressions, first round, last round, and semantic similarity



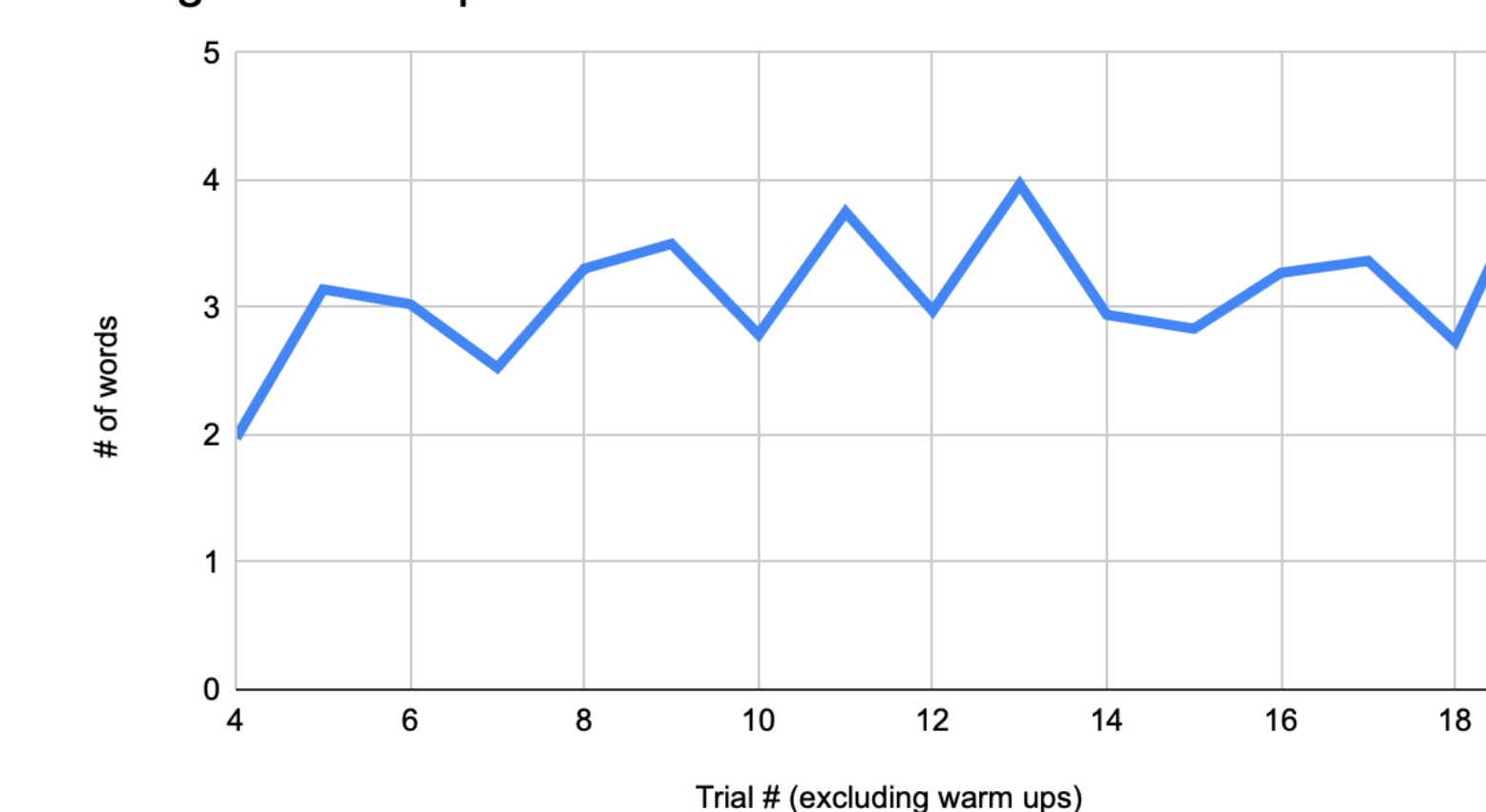
Results



Average response time over time



Length of descriptions over time



Discussion

- Unlike what was claimed in the 1966 study, we see that kids are capable of using ad hoc referential expression, and do so with high accuracy, and get faster over time like adults do.
- Linguistic skill acquisition does not necessarily need to come before skill employment

Acknowledgements

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